

TryHackMe — Computer Fundamentals Module

Computer Types

Learning Notes — All 8 Types, Trade-offs & Security Implications

"There is no best computer. There is only the right tool for the job."
— TryHackMe, Computer Types



1. Introduction — Sophia's Story

The TryHackMe Computer Types room follows Sophia, a summer intern at Nova Labs. Her journey of discovery — from connecting a smart fridge to her home Wi-Fi, through her first month learning about traditional computers, to her second month discovering hidden computers in everyday objects — is the narrative thread that ties all eight computer types together.

Sophia's Opening Insight

Sophia was trying to connect a new device to her home Wi-Fi when she noticed something unexpected: 'NexusCool Fridge X17'. She soon realised her neighbour had bought a smart refrigerator — a fridge with a built-in computer capable of connecting to the internet. Computers were no longer just laptops and phones. They were quietly infiltrating everyday objects, from kitchen appliances to doorbells. Her final report began: 'I arrived thinking computers had screens and keyboards. I leave knowing they are everywhere — especially where I do not see them.'

1.1 The Two Categories of Computer Interaction

The room divides computer types by how people interact with them — a practical framework that maps directly to how security professionals encounter different systems:

Category	Types	Characteristic
Direct-use computers	Laptop, Desktop, Workstation, Smartphone, Tablet	You physically interact with them — you see them, touch them, and are aware you are using a computer.
Indirect-use computers	Server, IoT, Embedded	They work in the background serving your requests. You rarely touch them directly and often don't know they exist.

2. The 8 Computer Types — Deep Dive

Every computer type reflects a specific set of design trade-offs. The same fundamental components — CPU, RAM, storage, network — appear in all of them, but in different configurations optimised for different priorities: portability, performance, reliability, size, power efficiency, or cost.



Laptop — Portable everyday computing

Screen & Keyboard: Yes — built-in screen and keyboard | *Story context: Sophia's first computer at Nova Labs — perfect for emails and documents, but slowed on long tasks due to cooling limits*

- Designed for portability: battery-powered, lightweight, all components integrated into one unit.
- Cooling is a fundamental challenge — thin chassis limits airflow, so thermal throttling limits sustained performance.
- Components are more compact and power-efficient than desktop equivalents — typically slower under sustained load.
- Integrated GPU in most consumer laptops (shared RAM); discrete GPU available in gaming/creator laptops.
- Battery life vs performance is the primary design trade-off — high performance drains battery faster.
- Common use: students, office workers, remote work, travel — any scenario requiring mobility.
- Modern laptops: Intel Core / AMD Ryzen processors; Apple M-series (ARM architecture) in MacBooks.

Real-world examples: MacBook Pro, Dell XPS, Lenovo ThinkPad, HP EliteBook, ASUS ROG (gaming).

Security relevance: Laptops are high-value theft targets — full of personal data. Drive encryption (BitLocker/FileVault) is critical. Unattended laptops in public spaces are vulnerable to Evil Maid attacks (physical access while owner is away). Theft of a laptop = theft of all unencrypted data on it.



Desktop — Sustained performance at a fixed location

Screen & Keyboard: Yes — external monitor, keyboard, mouse | *Story context: Sophia noticed the desktop stayed in one place, used wall power, and had better cooling — the same task ran smoothly much longer*

- Remains stationary — plugged into mains power permanently, enabling larger components and better cooling.
- Better thermal performance than laptops: larger chassis, multiple fans, optional liquid cooling, unrestricted airflow.
- Upgradeable — RAM, storage, GPU, and even CPU can often be replaced independently.
- No battery constraint — full power available at all times for sustained workloads.
- More cost-effective per unit of performance than laptops — you pay for hardware, not miniaturisation.
- Common use: office workstations, home computing, gaming PCs, small business servers.
- Tower, All-in-One (AIO), Mini-PC — different form factors for different space requirements.

Real-world examples: Dell OptiPlex, HP ProDesk, Apple iMac, custom-built gaming PCs.

Security relevance: Physical security is important — desktops are harder to steal than laptops but easier to access physically in shared offices. USB ports are attack vectors. Unlocked desktops in shared spaces allow direct OS access. Full-disk encryption and screen lock policies are essential.



Workstation — Precision and reliability for professional tasks

Screen & Keyboard: Yes — professional displays, keyboard, mouse | *Story context: Gabriel showed Sophia a workstation for simulations and 3D models — it looked like a desktop but was built for accuracy and reliability*

- Looks like a desktop but built to a higher standard — specialised components optimised for accuracy in long computations.
- Uses ECC RAM (Error-Correcting Code memory) — detects and corrects single-bit memory errors, preventing corrupted outputs in critical work.
- Higher-grade CPUs: often Intel Xeon or AMD Threadripper — more cores, larger cache, higher memory bandwidth than consumer chips.
- Professional GPUs: Nvidia Quadro/RTX A-series, AMD Radeon Pro — certified for CAD, simulation, and rendering software.
- Built for reliability in sustained, long-running workloads: 3D rendering, scientific simulation, video production, financial modelling.
- More expensive than equivalent consumer hardware — reliability and certification premium.
- Common use: architects, engineers, video editors, scientists, financial analysts, security researchers.

Real-world examples: HP Z-series, Dell Precision, Apple Mac Pro, Lenovo ThinkStation.

Security relevance: Workstations hold sensitive professional data — CAD files, financial models, research data. ECC RAM reduces the risk of corruption in critical computations. Physical and logical access controls are essential. Workstations in research or financial settings are high-value targets for industrial espionage.



Server — Providing services to many users over a network

Screen & Keyboard: No screen, no keyboard — headless operation | *Story context: Gabriel showed Sophia a room full of machines with no screens. They ran continuously, answering requests from multiple users simultaneously — powering every tool she used*

- Purpose-built to serve requests from many clients simultaneously — designed for continuous 24/7 operation.
- Headless: no monitor or keyboard attached — managed remotely via SSH, RDP, or web interfaces.
- Hardware redundancy: multiple PSUs (if one fails, the other takes over), RAID storage (multiple disks in fault-tolerant arrays), hot-swap components.
- Rack-mounted: installed in server racks in data centres — standardised 1U, 2U, 4U form factors.
- High RAM capacity: servers commonly have 64GB–1TB of RAM for handling many concurrent requests.
- Operating systems: typically Linux (Ubuntu Server, RHEL, CentOS) or Windows Server.
- Types: web servers, file servers, database servers, mail servers, DNS servers, application servers.

Real-world examples: Dell PowerEdge, HP ProLiant, Supermicro, AWS EC2 instances (cloud servers).

Security relevance: Servers are the primary targets in most cyber attacks — they hold databases, credentials, customer data, and source code. They are internet-facing, always on, and handle sensitive data. Common attacks: SSH brute force, web exploits, SQL injection, privilege escalation. Hardening: disable unused services, SSH key auth only, firewall rules, intrusion detection, regular patching.



Smartphone — The most powerful pocket-sized computer — the most popular computer most people own

Screen & Keyboard: Yes — touchscreen display | *Story context: Sophia's internship opened with the smart fridge discovery — but smartphones were already her most-used computer*

- The most popular type of personal computer globally — billions in use. The most powerful pocket-sized computer most people own.
- ARM processor architecture (Apple A-series, Qualcomm Snapdragon, Google Tensor) — highly power-efficient.
- All components integrated: CPU, GPU, cellular modem, Wi-Fi, Bluetooth, GPS, NFC, sensors — all in one SoC (System on Chip).
- Battery-powered: power efficiency is the dominant design constraint.
- Two primary ecosystems: Apple iOS (iPhone) and Google Android — different security models.
- App stores provide a managed distribution channel for software — with varying degrees of code review.
- Always connected: cellular (4G/5G), Wi-Fi, and Bluetooth simultaneously — large network attack surface.

Real-world examples: iPhone 16, Samsung Galaxy S25, Google Pixel 9, OnePlus 13.

Security relevance: Smartphones carry the most sensitive personal data of any device most people own: contacts, messages, emails, photos, banking apps, location history, health data. They are

always on and always connected. Risks: malicious apps, SMS phishing (smishing), public Wi-Fi attacks, physical theft, spyware. Mobile Device Management (MDM) is used by organisations to enforce security policies on employee smartphones.



Tablet — Touch-first experience with a larger screen than a smartphone

Screen & Keyboard: Yes — large touchscreen | *Story context: Tablets bridge the gap between smartphones and laptops — Sophia encountered them as shared devices at Nova Labs*

- Larger screen than a smartphone but more portable than a laptop — optimised for media consumption and touch-first interaction.
- Uses the same ARM-based SoC processors as smartphones — similar power efficiency.
- Runs mobile operating systems: iPadOS (Apple), Android (Google) — same app ecosystem as smartphones.
- Optional accessories: keyboard cover, stylus — transforms tablet into a lightweight laptop-like device.
- Limited by touch-first design for complex productivity tasks — keyboard and mouse support helps but is secondary.
- Common use: media consumption, education (schools), point-of-sale systems, field work, presentations.
- iPads used in enterprise for field workers, sales staff, and healthcare (clinical documentation).

Real-world examples: Apple iPad Pro, Samsung Galaxy Tab S9, Microsoft Surface Pro (hybrid tablet/laptop).

Security relevance: Tablets share the same security risks as smartphones — sensitive data, always connected, theft risk. Enterprise tablets used in POS systems or healthcare are high-value targets. Unsecured tablets in public-facing kiosks (hotels, airports, restaurants) are vulnerable to data exposure and app tampering.



IoT — Internet of Things — Single-purpose networked devices that connect and report

Screen & Keyboard: Sometimes — varies by device | *Story context: Sophia's neighbour's smart fridge — NexusCool Fridge X17 — was her first encounter with IoT; it connected to the network to receive commands*

- IoT devices are small, single-purpose computers that connect to a network to send data or receive commands.
- The defining characteristic vs embedded: connectivity — IoT devices are network-connected by design.
- Examples: smart doorbells, smart thermostats, smart fridges, smart TVs, security cameras, smart speakers, wearables.
- Typically run a lightweight embedded Linux or RTOS (Real-Time Operating System).
- Designed for long, unattended operation — often battery-powered for years.
- Communicate via Wi-Fi, Bluetooth, Zigbee, Z-Wave, LoRa, or cellular protocols.
- Managed via smartphone apps or cloud platforms — owner rarely interacts directly with the device hardware.

Real-world examples: Amazon Echo, Google Nest Thermostat, Ring Doorbell, Philips Hue lights, Fitbit, Samsung smart fridges.

Security relevance: IoT is one of the most critical security challenges today. Risks: default or unchanged passwords, infrequent patching, no encryption, weak authentication, large attack surface. Compromised IoT devices are recruited into botnets (Mirai botnet used IoT devices for massive DDoS attacks). Smart home devices can be used for surveillance, network intrusion, or lateral movement into the home network.



Embedded Computer — Invisible, reliable, single-purpose — inside the machine

Screen & Keyboard: No — no screen or user interface | *Story context: Sophia walked through automatic doors at Nova Labs every day, never realising a tiny computer was detecting her movement and signalling the motor — that is embedded computing*

- Embedded computers are built into a device to perform one specific function — often invisible to the user.
- The defining characteristic vs IoT: no network connectivity required — they do their job silently inside the machine.
- Examples: the computer in a car's ABS braking system, aircraft flight control, a coffee machine, a microwave, an automatic door, a pacemaker.
- Often run for years without any software update or human interaction.
- Extremely resource-constrained — tiny CPU, minimal RAM, very small storage, ultra-low power consumption.
- Reliability is paramount — embedded computers in safety-critical systems must operate without failure.
- Programmed in C or assembly; run a simple RTOS (FreeRTOS, VxWorks) or bare-metal firmware.

Real-world examples: Car ECUs (engine control units), ATM internals, medical device controllers, aircraft avionics, industrial PLCs.

Security relevance: Embedded systems are increasingly targeted by sophisticated attackers, especially in critical infrastructure and medical devices. Vulnerabilities are catastrophic: a compromised car brake system or pacemaker controller is life-threatening. Patching is extremely difficult — devices may be inaccessible or require hardware replacement. Stuxnet was an embedded systems attack targeting industrial PLCs (Programmable Logic Controllers).

3. The Master Comparison — All 8 Types at a Glance

Type	Screen?	Keyboard?	Primary Purpose	Key Design Priority
LAPTOP	Yes	Yes (built-in)	Portable everyday computing	Portability — battery life and compactness over raw performance
DESKTOP	Yes	Yes (external)	Sustained performance at fixed location	Performance & upgradeability — no battery or size constraint
WORKSTATION	Yes	Yes (external)	Precision tasks — 3D, simulation, analysis	Accuracy & reliability — ECC RAM, certified GPU, sustained compute
SERVER	No	No (headless)	Serve many users over a network 24/7	Availability & redundancy — multiple PSUs, RAID, always-on
SMARTPHONE	Yes	Yes (touch)	Pocket-sized personal computing & comms	Power efficiency — ARM SoC, battery life, always-connected
TABLET	Yes	Yes (touch)	Touch-first media & productivity	Screen size & portability — larger than phone, lighter than laptop
IOT	Sometimes	Rarely	Networked single-purpose sensing/control	Connectivity & efficiency — long unattended operation, low power
EMBEDDED	No	No	Silent single-purpose task inside a machine	Reliability — years of unattended operation, no updates needed

4. IoT vs Embedded — The Critical Distinction

The room emphasises this distinction because it is frequently confused, and it has major security implications. Both IoT and embedded devices are small and single-purpose — but connectivity is what separates them.

Feature	IoT Device	Embedded Computer
Connectivity	Connected to a network — it is the defining feature	May have no network connection at all
Examples	Smart doorbell, smart fridge, Fitbit, smart speaker	Car ABS controller, microwave timer, pacemaker, automatic door sensor
Updates	Can receive software updates over-the-air (OTA) via network	Firmware often never updated — or requires physical hardware access
Interaction	Controlled via smartphone app or cloud platform	Operates autonomously — no user interaction needed or expected
Lifetime	Few years — technology evolves, users replace devices	Decades — embedded in machinery with 20-30 year operational life
Security risk	Internet-facing: can be remotely attacked, recruited to botnets	Physical access required to attack — but catastrophic if compromised
Room answer	"IoT devices connect to a network to report data or receive commands"	"Embedded computers might not connect to anything; they do their job inside the machine"

The Room's Key IoT vs Embedded Quote

Both can be small and single-purpose. The difference is connectivity. IoT devices connect to a network to report data or receive commands. Embedded computers might not connect to anything — they do their job inside the machine, often for years without anyone knowing they exist. That's embedded computing: invisible, reliable, everywhere.

5. The Three Core Design Trade-offs

The room concludes with an important insight: there is no best computer. Every design involves trade-offs, and the right choice depends entirely on the intended purpose. Three fundamental trade-offs appear repeatedly:

Trade-off	What You Gain	What You Sacrifice
Mobility vs Performance	Portability — carry your computer anywhere, use on battery	Sustained performance — cooling is harder, thermal throttling limits peak output under load
Reliability vs Cost	Redundancy — extra PSUs, ECC RAM, RAID storage, certified components	Budget — server-grade and workstation hardware costs significantly more than consumer equivalents
Purpose vs Flexibility	Optimisation — a device built for one task does that task extremely well	Versatility — a single-purpose device cannot easily perform other tasks

The Three Trade-off Lessons from the Room

Mobility costs power — smaller, portable computers must sacrifice sustained performance. Reliability costs money — servers and critical systems use redundancy (extra PSUs, RAID disks) to avoid failure. Purpose shapes everything — you touch a phone; you ask a server for information; an IoT device works quietly without demanding attention.

5.1 The 'Right Tool for the Job' Framework

When choosing a computer type, ask three questions:

Question	Answer → Type	Example
Does it need to move?	Yes → Laptop/Phone/Tablet; No → Desktop/Server/Workstation	Field engineer → laptop; Data centre → rack server
Will it serve multiple users?	Yes → Server; No → Personal computer types	Web application → server; Individual productivity → desktop/laptop
Is it embedded in a product?	Yes + networked → IoT; Yes + standalone → Embedded; No → personal/server	Smart thermostat → IoT; Car ECU → embedded; Office work → desktop
Does it need high reliability?	Yes → Server/Workstation (ECC, redundancy); No → consumer devices	Medical imaging → workstation; Home gaming → consumer desktop
Is power consumption critical?	Yes → Smartphone/Tablet/IoT/Embedded; No → Desktop/Server	Wearable → ARM SoC; On-premises server → high-performance CPU

6. Room Answers & Scenario Examples

6.1 Scenario Classification — Which Type Is It?

Scenario	Computer Type
Sophia uses it for emails and documents while at her desk at Nova Labs	Laptop
A machine that stayed in one place, used wall power, and had better cooling	Desktop
A machine built for 3D models and simulations — prioritises accuracy	Workstation
Machines in a room with no screens — running continuously, serving tools	Server
The most powerful pocket-sized computer and the most popular type people own	Smartphone
Sophia's neighbour's NexusCool Fridge X17 — connects to the internet	IoT
The computer inside the automatic door at Nova Labs — detects movement	Embedded
A larger touch-screen device between phone and laptop in size	Tablet
Keeps doors opening, planes flying, and coffee machines brewing silently	Embedded
Can receive software updates over-the-air; controlled via smartphone app	IoT

7. Security Implications — The Attack Surface of Each Type

Every computer type presents a different security attack surface. A security professional must understand not just what a computer type is, but why it presents distinct risks and how those risks are addressed.

Type	Primary Security Risk	Key Defences
LAPTOP	Physical theft — all local data exposed if unencrypted	Full-disk encryption (BitLocker/FileVault), strong login password, remote wipe capability
DESKTOP	Physical access in shared spaces — USB attack vectors	USB port controls, screen lock policy, FDE, physical access restriction
WORKSTATION	Sensitive professional data — industrial espionage target	Access controls, data classification, network segmentation, regular backups
SERVER	Internet-facing, always on — primary target for all attacks	Hardening, SSH keys only, firewall, IDS/IPS, patch management, least privilege
SMARTPHONE	Constant connectivity — personal data, banking, location	MDM policies, screen lock, app permissions review, avoid public Wi-Fi or use VPN
TABLET	Shared device exposure — kiosk and POS terminal tampering	MDM, kiosk mode, screen lock, regular app audits
IOT	Default credentials, no patching, botnet recruitment (Mirai)	Change default passwords, VLAN isolation, firmware updates, disable unused features
EMBEDDED	Life-safety impact — car, medical, industrial control compromise	Supply chain security, code signing, hardware security modules, physical protection

8. Key Takeaways & Glossary

8.1 Key Takeaways

- There are 8 computer types: Laptop, Desktop, Workstation, Server, Smartphone, Tablet, IoT, Embedded.
- Computers are divided into direct-use (you interact with them) and indirect-use (they work in the background).
- The most popular pocket-sized computer (most people own) is: Smartphone.
- Laptop: portable, battery-powered, thermally limited under sustained load.
- Desktop: stationary, mains-powered, better cooling, upgradeable, more performance per pound.
- Workstation: looks like a desktop but has ECC RAM, certified GPU, designed for accuracy in long computations.
- Server: no screen or keyboard, runs 24/7, serves many users, uses hardware redundancy (multi-PSU, RAID).
- IoT devices **CONNECT** to a network; embedded computers may **NOT** — this is the critical distinction.
- IoT examples: smart fridge, smart doorbell, smart thermostat, fitness tracker.
- Embedded examples: automatic door sensor, car ABS, pacemaker, coffee machine timer, aircraft avionics.
- The three trade-offs: Mobility vs Performance, Reliability vs Cost, Purpose vs Flexibility.
- There is no 'best' computer — only the right tool for the job.
- Sophia's closing quote: 'I arrived thinking computers had screens and keyboards. I leave knowing they are everywhere — especially where I do not see them.'

8.2 Core Glossary

Term	Definition
Laptop	Portable computer with integrated screen, keyboard, and battery — trades sustained performance for mobility.
Desktop	Stationary computer using mains power — better cooling and performance than laptop; not portable.
Workstation	High-spec computer for professional tasks — ECC RAM, certified GPU, built for accuracy and reliability.
ECC RAM	Error-Correcting Code RAM — detects and corrects single-bit memory errors. Used in servers and workstations.
Server	Headless computer (no screen/keyboard) running 24/7 to serve many users — uses hardware redundancy.
Headless	A computer with no monitor or keyboard attached — managed entirely via remote access (SSH, RDP).

RAID	Redundant Array of Independent Disks — multiple disks providing fault tolerance if one fails.
Smartphone	The most popular pocket-sized computer — ARM SoC, always connected, touch interface, battery-powered.
ARM	Processor architecture used in smartphones, tablets, and embedded devices — power-efficient vs x86.
SoC	System on Chip — integrates CPU, GPU, modem, and other components onto a single chip. Used in mobile devices.
Tablet	Touch-first device larger than a smartphone — optimised for media consumption and portable productivity.
IoT	Internet of Things — single-purpose networked devices (smart home, wearables, connected appliances).
Embedded Computer	A computer built inside a device to perform one specific task — may have no network connection.
RTOS	Real-Time Operating System — lightweight OS for embedded systems requiring precise, deterministic timing.
Botnet	A network of compromised devices (often IoT) controlled by an attacker — used for DDoS, spam, mining.
Mirai	Famous IoT botnet malware that compromised millions of IoT devices using default credentials for DDoS attacks.
MDM	Mobile Device Management — software for managing and enforcing security policies on smartphones and tablets.
Full Disk Encryption	Encrypts entire storage — essential on laptops to protect data if stolen.
Trade-off	A design compromise — gaining one quality requires sacrificing another (portability vs performance, etc.).
Direct-use	Computers you physically interact with — laptops, desktops, smartphones, tablets.
Indirect-use	Computers working in the background serving your requests — servers, IoT, embedded.

8.3 Next Steps on TryHackMe

- **Immediate:** Continue with the Operating Systems: Introduction room — how the OS layers above the hardware covered here.
- **Foundation:** Complete the Windows Fundamentals rooms — the OS running on desktops, workstations, and servers.
- **Foundation:** Study the Linux Fundamentals rooms — the OS used on most servers and IoT devices.

- **Security:** Take the IoT Security room — deep-dive into securing connected devices and the unique challenges they present.
- **Advanced:** Study the Embedded Systems Security path — for critical infrastructure and ICS/SCADA security.

Computers Are Everywhere — Especially Where You Don't See Them

The most important insight from this room is that computing is not limited to devices with screens and keyboards. Servers power the internet. IoT devices populate our homes and cities. Embedded computers keep planes in the air and regulate medical devices. As a security professional, you will need to protect all of them — each with different tools, techniques, and threat models.

Source: TryHackMe — Computer Types Room | tryhackme.com/room/computertypes